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Radio-Frequency-Based Unmanned-Aerial-Vehicle Identification Using Cyclic-Paw-Print

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***Abstract*—A novel robust unmanned aerial vehicle (UAV) identification approach using our recently proposed cyclic-paw-print (CPP) features extracted from the radio-frequency (RF) signals is presented in this paper for air-traffic surveillance and control. In this work, a radio-frequency (RF) signal such as a UAV telemetry, track, and command (TT&C) signal and a digital data-transmission (DDT) signal emitted by a UAV is first received and segmented according to its short-time energy-to-spectral-entropy ratio (ST-ESER). Next, cyclic-spectrum analysis is applied to each selected RF signal segment to form the corresponding polyspectrum, which is further utilized to establish the cyclic-paw-print (CPP) matrices. Finally, the lightweight convolutional neural network, namely ResNet-18, which takes the produced CPPs as the input features, is trained to recognize the aforementioned RF signals so the corresponding UAVs can be identified. Monte Carlo simulation results demonstrate the superior effectiveness of our proposed new UAV identification approach in comparison with other existing deep-learning methods.**

***Index Terms*—Unmanned aerial vehicle (UAV), radio-frequency (RF) based UAV identification, short-time energy-to-spectral-entropy ratio (ST-ESER), cyclic-paw-print (CPP), lightweight convolutional neural network.**

I. Introduction

Nowadays, unmanned aerial vehicles (UAVs), which exhibit many advantages in practice, are very important in various civil and military applications, including 6G (sixth generation) communication networks, precision agriculture, logistic services, traffic monitoring, law-enforcement surveillance, military reconnaissance, and precision attack [1], [2]. However, abuse of UAVs would definitely draw concerns on safety, privacy, and security, where UAVs may be used, more and more often, for privacy infringement, terrorist attacks, and cyber attacks [3], [4]. Meanwhile, the increasing use of UAVs poses serious challenges to UAV regulation and air-traffic control (ATC) worldwide. In particular, *low-altitude, small, slow (LSS) UAVs* are fairly difficult to identify and control in a complex urban environment [5]. Because UAV surveillance and patrol become crucial, robust UAV identification techniques are in urgent demand.

Currently, existing UAV detection and identification methods mostly rely on radar, acoustics, imaging, and radio-frequency (RF) reception [6]. Although radar has become a promising means to detect and monitor large and medium aerial targets such as aircraft, airships, missiles, and rockets, it is much less reliable for identifications of LSS UAVs as limited by their tiny radar cross-sections. The S/X/Ku-band high-power radars could enhance the recognition accuracy of LSS UAVs but the required cost is, generally speaking, unbearably high [7]. Besides, the micro-Doppler effect can be utilized for UAV identification but the pertinent approach can only work in the high signal-to-noise ratio (SNR) scenarios [8]. Like all other sound source-localization schemes, acoustic signatures can be extracted from UAVs to distinguish them. Nonetheless, it is well known that the acoustic reception quality is limited by the sensing distance and the identification performance significantly deteriorates in the presence of (noticeable) ambient noise [9], [10]. Image-based UAV sensing methods rely on high-definition photos acquired by surveillance cameras, which often suffer from poor illumination, bad weather conditions, and visual obstruction [11], [12]. On the other hand, RF-based UAV identification methods sense the RF signals transmitted during the communications between a UAV and

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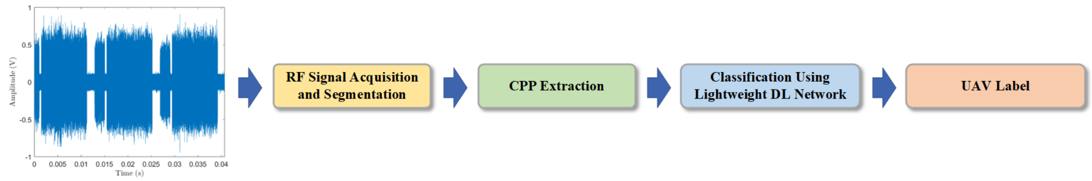


Fig. 1. The block diagram of our proposed novel RF-based UAV identification approach.

its controller using high-gain antennae and high-sensitivity receivers in both light-of-sight (LOS) and non-line-of-sight (NLOS) scenarios [11]. Then, the essential RF fingerprints are extracted from the captured RF signals for UAV identification. RF-based UAV identification methods can lead to outstanding recognition accuracies at low cost and they are robust against environmental variations [9]. Therefore, we would like to investigate a novel RF-based UAV detection and identification approach in this work.

According to our recent study of automatic composite-modulation classification for the TT&C signals emitted by UAVs, a robust RF characteristic (fingerprint), namely *cyclic-paw-print* (CPP), can be extracted from the normalized second-order cyclic spectrum (NSCS) of a received signal [13]. With the help of the aforementioned new CPP features, we design a novel RF-based UAV identification scheme using a lightweight deep-learning (DL) network, namely ResNet-18 in this work [14]. In our proposed new UAV identification approach, the acquired RF signal is segmented and those segments with high *fidelity* are further selected according to the associated *short-time energy-to-spectral-entropy ratio* (ST-ESER). Next, the *cyclic-spectrum analysis* is employed over the selected (high-fidelity) segments to build the CPP matrices. These CPP matrices will be utilized as the input features of ResNet-18 to identify the UAV. To evaluate the performance of our proposed RF-based UAV identification approach in comparison with other existing UAV recognition methods, we carry out numerous simulations using the DroneRFa dataset in [15].

The rest of this paper is organized as follows. Section II presents the details of our proposed novel RF-based UAV identification scheme. Section III demonstrates performance evaluation and comparison. Conclusion will be drawn finally in Section IV.

II. Our Proposed Novel UAV Identification Scheme

The system diagram of our proposed novel UAV identification approach is illustrated by Figure 1, which consists of three major mechanisms, namely (*i*) *RF signal acquisition and segmentation*, (*ii*) *CPP extraction*, and (*iii*) *classification using lightweight DL network*. Without loss of generality, we assume that the UAV candidate set $\mathbb{K} \stackrel{\text{def}}{=} \{\mathscr{K}_1, \mathscr{K}_2, \cdots, \mathscr{K}_K\}$ is composed by K types of RF signals, where the k-th type of RF signal is labeled by $\mathscr{K}_k$, $k = 1, 2, \ldots, K$. Details of the aforementioned three mechanisms in Figure 1 will be presented in the following subsections.

A. RF Signal Acquisition and Segmentation

According to Figure 1, one can continuously acquire RF signals within the band of UAV operation using a monitoring receiver. According to a captured RF signal waveform illustrated by Figure 2, we can find that the communications between a UAV and its controller do not always remain active [16]. Furthermore, according to Figure 2, the acquired RF signal waveform can be split into "*image-transmission intervals* (ITIs)", "*silent intervals* (SIs)", and "*frequency-hopping intervals* (FHIs)". Note that we can only extract the features from the ITIs and FHIs for UAV identification. Therefore, we need to design a sophisticated mechanism for detection and segmentation of captured RF signals in this work.

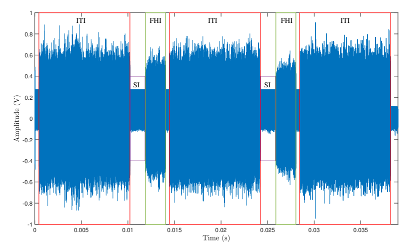


Fig. 2. Illustration of a captured RF signal waveform produced by a UAV (DJI Phantom 4 Pro) and its controller. The signal data are acquired from the DroneRFa dataset in [15].

Let's express a capture discrete-time RF signal sequence by $r(n)$, n=0, 1, …, $M-1$ where M denotes the sample size. We further use a sliding window (frame) to segment $r(n)$ such that the i-th frame of signal samples can be represented by $\zeta_i(n)$, n=0, 1, …, $\mathscr{N}-1$ where

$$\zeta_i(n) \stackrel{\text{def}}{=} r\Big(n + (i-1)\mathscr{L}\Big), \; n = 0, 1, \ldots, \mathscr{N} - 1, \quad (1)$$

where i=1, 2, …, $\mathscr{I}$ and $\mathscr{N}-\mathscr{L}$ denotes the number of overlapped samples of two consecutive frames i and i+1. Note that

$$\mathscr{I} \stackrel{\text{def}}{=} \left\lfloor \frac{M - \mathscr{N} + \mathscr{L}}{\mathscr{L}} \right\rfloor, \quad (2)$$

where "$\lfloor\ \rfloor$" denotes the integer-rounding-down operator. According to Eqs. (1) and (2), one can divide $r(n)$ into the frames $\zeta_i(n)$'s with equal frame-length $\mathscr{N}$ as depicted by Figure 3. The discrete Fourier transform (DFT) sequence of each frame of signal samples $\zeta_i(n)$ for i=1, 2, …, $\mathscr{I}$ is thus given by

$$U_i(m) \stackrel{\text{def}}{=} \sum_{n=0}^{\mathscr{N}-1} \zeta_i(n) \times \exp\left(-\frac{\iota 2\pi n m}{\mathscr{N}}\right), \; m = 0, 1, \ldots, \mathscr{N}, \quad (3)$$

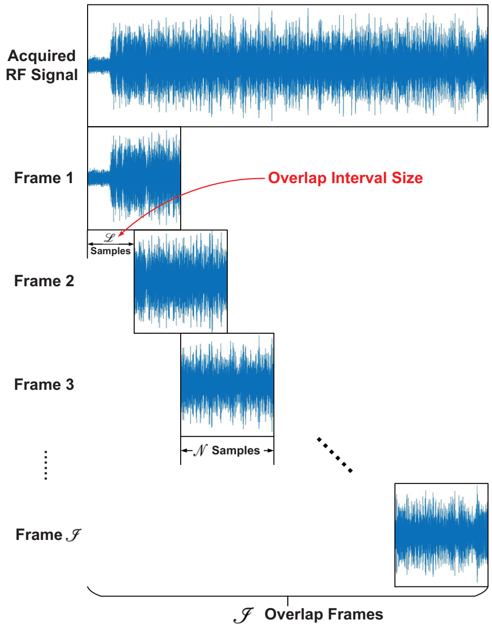


Fig. 3. Illustration of a captured discrete-time RF signal sequence $r(n)$ and the frames $\zeta_i(n)$'s.

where $\iota \stackrel{\text{def}}{=} \sqrt{-1}$ and i=1, 2, ..., $\mathscr{I}$. According to Eq. (3), we can also obtain the *magnitude spectrum sequence* of the i-th frame as $|U_i(m)|$, m=0, 1, ..., $\mathscr{N}$. According to [17], we can convert $|U_i(m)|^2$ to a discrete probability sequence $p_i(m)$ such that

$$p_i(m) \stackrel{\text{def}}{=} \frac{|U_i(m)|^2}{\sum_{m=0}^{\mathscr{N}-1} |U_i(m)|^2}, \quad m = 0, 1, \ldots, \mathscr{N}-1. \tag{4}$$

We call the probability sequence $p_i(m)$ formulated by Eq. (4) the "*spectral probability sequence*". Consequently, we can find the corresponding entropy, namely the "*spectral entropy*", as given by

$$H_i \stackrel{\text{def}}{=} -\sum_{m=0}^{\mathscr{N}-1} p_i(m) \times \ln\left[p_i(m)\right]. \tag{5}$$

Let's define an ST-ESER of the i-th frame by

$$\rho_i \stackrel{\text{def}}{=} \frac{\sum_{m=0}^{\mathscr{N}-1} |U_i(m)|^2}{H_i}, \tag{6}$$

where H_i is given by Eq. (5). Note that the numerator (the frame signal energy) and the denominator (the spectral entropy) of ρ_i will become large and small, respectively, in the presence of a UAV signal. As a result, the larger the ST-ESER ρ_i, the more confident we are that a UAV signal does appear in the i-th frame. In other words, the frame i can be selected according to the following decision-metric:

$$\begin{cases} \text{Frame } i \text{ is selected} & \text{if } \rho_i \geq \hbar, \\ \text{Frame } i \text{ is discarded} & \text{if } \rho_i < \hbar, \end{cases} \tag{7}$$

where the threshold $\hbar$ can be determined according to

$$\hbar \stackrel{\text{def}}{=} \frac{\varsigma}{\mathscr{I}} \sum_{i=1}^{\mathscr{I}} \rho_i \tag{8}$$

and the factor ς can be determined empirically (it is suggested that $0.5 \leq \varsigma \leq 0.9$). After each frame i is tested using Eq. (7), we can aggregate the consecutive selected frame signal sequences to form a segment of selected signal sequence. For example, let the ν consecutive frames: i', $i'+1$, ..., $i'+\nu-1$ all be selected frames according to Eq. (7). Then the corresponding entire segment of selected signal sequence can be written as Eq. (9) on top of Page 4. We assume that ITIs and FHIs do not overlap with each other. Note that the length of a segment consisting of ITIs is usually much larger than that of a segment consisting of FHIs as illustrated by Figure 2. Thus, we can easily judge whether a segment of aggregated selected signal sequence $x(n)$ given by Eq. (9) is a segment consisting of ITIs or FHIs.

B. CPP Extraction

The second-order cyclic spectrum (SCS) is proposed here to characterize the received UAV RF (control and communication) signals. The FFT (Fast Fourier Transform) accumulation method (FAM) is adopted to calculate the second-order cyclic-autocorrelation function of a discrete-time UAV RF signal sequence $x(n)$ according to [18], [19]. Furthermore, according to [18], [19], the time-smooth-based cyclic periodogram $\mathscr{S}_{x_T}^{\varepsilon}(n,f)_{\Delta t}$ for a spectral-frequency f and a cyclic-frequency ε can be calculated as

$$\mathscr{S}_{x_T}^{\varepsilon}(n,f)_{\Delta t} \stackrel{\text{def}}{=} \sum_{\lambda} X_T(\lambda, f_1)\, X_T^*(\lambda, f_2)\, g'(n-\lambda), \tag{10}$$

where $g'(n)$ is a unity-area weighting function of width $\Delta t \stackrel{\text{def}}{=} N T_s$ seconds, T_s is the sampling period, N denotes the number of samples per FFT window, f_1 and f_2 denote the center frequencies of the filters used in the FAM, i.e., $f_1 \stackrel{\text{def}}{=} f+\varepsilon/2$, $f_2 \stackrel{\text{def}}{=} f-\varepsilon/2$, and $X_T(\lambda, f_1)$, $X_T(\lambda, f_2)$ are the complex demodulates of $x(n)$, which can be obtained as

$$X_T(n,f) \stackrel{\text{def}}{=} \sum_{\lambda=-N'/2}^{N'/2-1} \hat{w}(\lambda)\, x(n-\lambda)\, e^{-\jmath 2\pi f (n-\lambda) T_s}. \tag{11}$$

Note that $\hat{w}(\lambda)$ denotes a data-tapering window of duration T=$N' T_s$ seconds and its bandwidth coincides with the frequency resolution Δf=f_s/N' of the SCS where f_s is the sampling frequency. Thus, the SCS can be estimated from the time-smoothed cyclic periodogram $\mathscr{S}_{x_T}^{\varepsilon}(n,f)_{\Delta t}$ given by Eq. (10) without bias such that

$$\mathscr{S}_{x}^{\varepsilon}(f) \stackrel{\text{def}}{=} \lim_{\Delta f \to 0} \lim_{\Delta t \to 0} \mathscr{S}_{x_T}^{\varepsilon}(n,f)_{\Delta t}. \tag{12}$$

Furthermore, according to Eq. (10), the normalized second-order cyclic magnitude spectrum (NSCMS) $\left|\overline{\mathscr{S}}_{x}^{\varepsilon}(f)\right|$ can be obtained as

$$\left|\overline{\mathscr{S}}_{x}^{\varepsilon}(f)\right| \stackrel{\text{def}}{=} \frac{|\mathscr{S}_{x}^{\varepsilon}(f)|}{\max_{\varepsilon, f} |\mathscr{S}_{x}^{\varepsilon}(f)|}. \tag{13}$$

$$x(n) \overset{\text{def}}{=} \begin{cases} \zeta_{i'}(n), & n = 0, 1, \ldots, \mathscr{N} - 1, \\ \zeta_{i'+1}(n - \mathscr{L}), & n = \mathscr{N}, \mathscr{N} + 1, \ldots, \mathscr{N} + \mathscr{L} - 1, \\ \vdots & \vdots \\ \zeta_{i'+\nu-1}\big(n - (\nu - 1)\mathscr{L}\big), & n = (\nu - 1)\mathscr{L} + \mathscr{N} - \mathscr{L}, (\nu - 1)\mathscr{L} + \mathscr{N} - \mathscr{L} + 1, \ldots, \nu\mathscr{L} + \mathscr{N} - \mathscr{L} - 1. \end{cases} \tag{9}$$

The NSCMSs of the UAV RF signals presented in [15] and their top views are depicted by Figure 4.

The top view of the NSCMS of a UAV RF signal would often be concentrated in the four symmetric regions of the $\varepsilon - f$ plane, which can be called a "*cyclic-paw-print* (CPP)" [19]. According to Figure 4, it is conspicuous that the CPPs resulting from different UAV RF signals are distinguishable. The NSCMS can be treated as a colormap and converted to a "quantized" image matrix $\tilde{\mathscr{S}}' \overset{\text{def}}{=} [\gamma_{u,v}]_{1 \le u \le 2N+1, 1 \le v \le N'+1}$ by a b-bit quantizer such as

$$\gamma_{u,v} \overset{\text{def}}{=} \left\lfloor 2^{b-1} \times \left| \overline{\mathscr{S}}_x^{\varepsilon_u}(f_v) \right| \right\rfloor, \quad u = 1, 2, \ldots, 2N + 1 \text{ and } v = 1, 2, \ldots, N' + 1, \tag{14}$$

where b is a positive integer and $\lfloor \ \rfloor$ denotes the floor function which outputs the greatest integer less than or equal to the real input number. Without loss of generality, b can be set to 16 in this work. The rectangular grayscale image $\tilde{\mathscr{S}}'$ is reshaped into a $\Lambda \times \Lambda$ square matrix and further transformed into a pseudo-color square image [13], namely the "*parula colormap*" using the corresponding function in MATLAB such that

$$\gamma_{\wp,q}^{\dagger} \xrightarrow{\text{parula}} \vec{\mathscr{R}}_{\wp,q}^{\text{RGB}}, \quad \wp = 1, 2, \ldots, \Lambda \text{ and } q = 1, 2, \ldots, \Lambda, \tag{15}$$

where

$$\vec{\mathscr{R}}_{\wp,q}^{\text{RGB}} \overset{\text{def}}{=} \left[\mathscr{R}_{\wp,q}^{\text{R}}, \mathscr{R}_{\wp,q}^{\text{G}}, \mathscr{R}_{\wp,q}^{\text{B}}\right]^T, \quad \wp = 1, 2, \ldots, \Lambda \text{ and } q = 1, 2, \ldots, \Lambda. \tag{16}$$

According to Eqs. (15) and (16), three square matrices can be obtained for the individual redness (R), greenness (G), and blueness (B) channels such that

$$\tilde{\mathscr{R}}_{\text{R}} \overset{\text{def}}{=} \left[\mathscr{R}_{\wp,q}^{\text{R}}\right]_{1 \le \wp \le \Lambda, 1 \le q \le \Lambda}, \tag{17}$$

$$\tilde{\mathscr{R}}_{\text{G}} \overset{\text{def}}{=} \left[\mathscr{R}_{\wp,q}^{\text{G}}\right]_{1 \le \wp \le \Lambda, 1 \le q \le \Lambda}, \tag{18}$$

$$\tilde{\mathscr{R}}_{\text{B}} \overset{\text{def}}{=} \left[\mathscr{R}_{\wp,q}^{\text{B}}\right]_{1 \le \wp \le \Lambda, 1 \le q \le \Lambda}. \tag{19}$$

Consequently, the CPP, which is the square pseudo-color image, can be represented by the three-order tensor:

$$\mathscr{R}_{\text{RGB}} \overset{\text{def}}{=} \left[\tilde{\mathscr{R}}_{\text{R}}, \tilde{\mathscr{R}}_{\text{G}}, \tilde{\mathscr{R}}_{\text{B}}\right]. \tag{20}$$

C. Classification Using Lightweight DL Network

The lightweight convolutional neural network, namely ResNet-18, is adopted here to identify the received UAV RF signal according to the extracted CPPs. ResNet-18, which includes the critical residual units to avoid gradient explosion, can provide a fairly fast convergence-speed and lead to a promising classification accuracy subject to a moderate network depth (a relatively simple architecture) [14]. In this work, the ResNet-18 model consists of a 7×7 convolutional layer, two max pooling layers, eight residual units, and a fully-connected layer, where each residual unit contains two 3×3 convolutional layers and a fully-connected layer. The CPP $\mathscr{R}_{\text{RGB}}$ is directly fed into the ResNet-18 model and the output of the last fully-connected layer is a K-dimensional feature vector $\vec{Z} \overset{\text{def}}{=} [z_1, z_2, \cdots, z_K]$. Finally, a softmax activation function as defined in [20], [21] can be utilized to identify the type of the received UAV RF signal such that

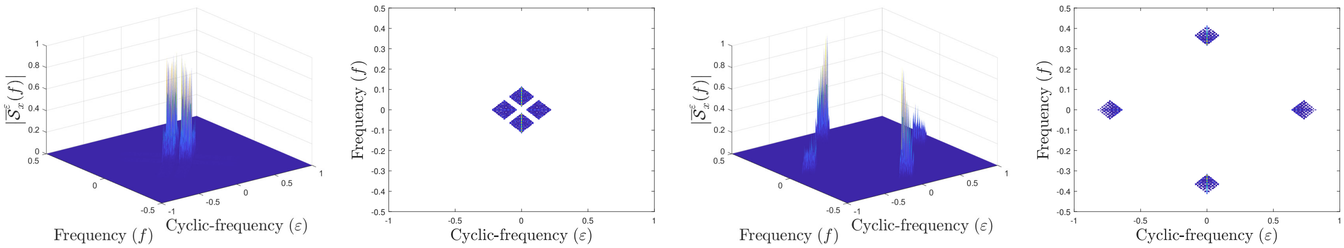
$$\mathscr{P}_k = \text{softmax}(z_k), \quad k = 1, 2, \ldots, K, \tag{21}$$

where $\mathscr{P}_k$ corresponds to the estimated *a priori* probability of the k^{th} type of UAV RF signal, namely $\mathscr{K}_k$ in $\mathbb{K}$, i.e., $\sum_{k=1}^{K} \mathscr{P}_k = 1$. Consequently, the type of the received UAV RF signal can be ultimately identified as $\mathscr{K}_{\hat{k}}$ where

$$\hat{k} \overset{\text{def}}{=} \underset{k \in \{1,2,\cdots,K\}}{\text{argmax}} \ \mathscr{P}_k. \tag{22}$$

III. Simulation

The performance of our proposed novel RF-based UAV identification approach is evaluated via Monte Carlo simulations in terms of *probability of correct classification* (P_{cc}) over the entire candidate set $\mathbb{K}$ versus the average signal-to-noise ratio (SNR) using the publicly available large-scale drone RF signal dataset (DroneRFa) in [15] and compared with that of the existing short-time Fourier transform (STFT) based UAV recognition method proposed in [15]. The DroneRFa dataset in [15] includes twenty-four types of RF communication signals between drones and their flight controllers as well as a type of background signal without drones, which covers small and medium-sized drones in common commercial and civilian fields. Here, all UAV RF communication signals in DroneRFa are regarded as the "noise-free" (clean) signals, and the additive Gaussian white noise (AWGN) is artificially added to control the corresponding SNRs ranging from -10 dB to 15 dB. During the training stage, $M = 10^7$ signal samples of each type of UAV RF signal in DroneRFa are collected and split into frames, where $\mathscr{N} = 10000$ and $\mathscr{L} = 2500$. Once the UAV RF signal subsequences within these frames are obtained, CPP images are constructed for all these frame. The FFT window size for generating the SCS in the FAM is fixed to $N' = 256$ while the sample size is set to be $N = 16384$. Note that $\tilde{\mathscr{S}}'$ is a 32769×257 matrix, which is further transformed into a $224 \times 224 \times 3$ tensor $\mathscr{R}_{\text{RGB}}$. The constructed CPP feature tensors $\mathscr{R}_{\text{RGB}}$ from all frames are then randomly divided into the training and validation subsets subject to the training-to-validation ratio of 7:3. Adam optimizer is utilized during

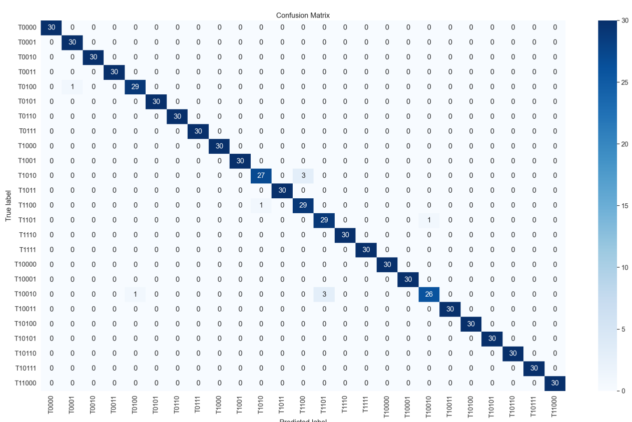


(a) NSCMS of DJI Phantom 4 Pro (b) Top view of (a) (c) NSCMS of DJI MATRICE 600 Pro (d) Top view of (c)

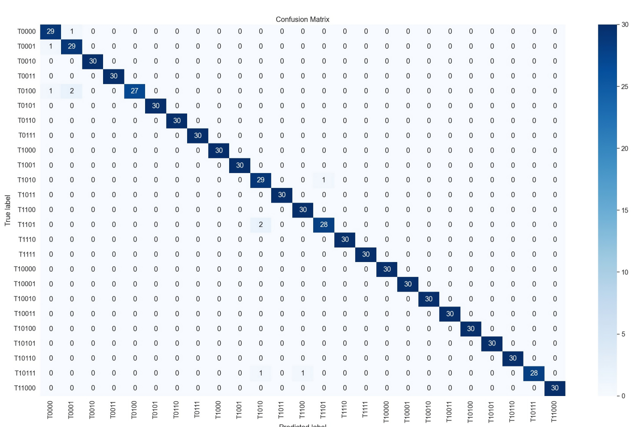
Fig. 4. The NSCMSs and their top views (CPPs) of UAV RF signals presented in [15].

training while the learning rate for updating each parameter is fixed to 0.001. The mini-batch size is set to be 32 to accelerate the convergence and improve the computational efficiency.

The confusion matrix of noise-free UAV identification resulting from our proposed new UAV identification approach and the existing STFT-based method in [15] are exhibited by Figure 5. It is obvious from Figure 5 that both methods cannot perfectly recognize all types of UAV RF signals provided in DroneRFa, but they can achieve the same P_{cc} of 98.7%. Then, AWGN is added to the UAV RF signals in the DroneRFa dataset to further evaluate the P_{cc}'s of our proposed new approach (denoted by "Our Method" in the figure) with respect to various average SNRs (from -10 to $15\,$dB). The results from our proposed new scheme are compared with those from the existing STFT-based method in [15] (denoted by "STFT" in the figure) as delineated by Figure 6. According to Figure 6, the P_{cc}'s produced by our proposed new scheme are always larger than or equal to those produced by the existing STFT-based method, and the P_{cc}'s resulting from both methods can reach up to 98% when SNR$\geq$10$\,$dB. Thus, our proposed new CPP-based approach obviously outperforms the existing STFT-based method in [15].



(a) Our Proposed New CPP-based Scheme



(b) The Existing STFT-based Scheme

Fig. 5. The confusion matrices of noise-free UAV identification resulting from our proposed new CPP-based UAV identification approach and the existing STFT-based method in [15]. The DroneRFa dataset in [15] is utilized.

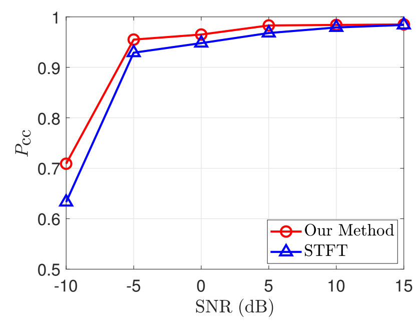


Fig. 6. Probabilities of correct classification P_{cc}'s versus average SNR η for recognizing UAV RF signals in DroneRFa resulting from our proposed new method using CPPs as the input features and the existing method in [15] using STFTs as the input features. The ResNet-18 model is adopted for both methods.

Meanwhile, to justify ResNet-18's advantage for CPP features over other DL models, the constructed CPPs of the UAV RF signals in the DroneRFa dataset are fed into three DL models, namely ResNet-18, ShuffleNet-V2 in [22], and Vision Transformer in [23], to identify UAV types under different average SNRs, respectively. The simulation parameters remain unchanged as stated before. The corresponding P_{cc}'s are evaluated and depicted by Figure 7. According to Figure 7, ResNet-18 adopted by our proposed UAV identification method can lead to the best recognition accuracy in comparison with ShuffleNet-V2 and Vision Transformer when the input features are CPPs. However, the overall recognition accuracy of Vision Transformer is relatively small across all average SNRs. When the average SNR reaches up to 5$\,$dB, the recognition accuracies

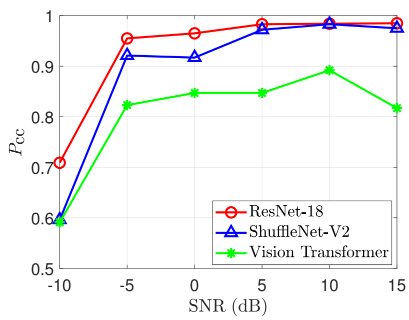


Fig. 7. Probabilities of correct classification P_{cc}'s versus average SNR η for recognizing UAV RF signals in DroneRFa resulting from three DL models. The input features are CPPs for all these three models.

of ShuffleNet-V2 and ResNet-18 are roughly the same and reach around 97%. According to [23], Vision Transformer is good at capturing long-range dependencies between patches but often disregards the local feature extraction as the two-dimensional patch is projected onto a vector by a simple linear layer. Let's further clarify such a drawback here. According to Figure 4, the distinction of the CPP features produced from different UAV RF signals relies on the "*high-intensity patterns*" (the dark blue regions). Vision Transformer will always divide such a CPP image into patches and project these patches onto a vector; it would likely result in loss of such high-intensity patterns in local patches and degrade the recognition performance. As a result, ResNet-18 is most suitable for our proposed CPP features.

IV. Conclusion

In this work, a novel RF-based UAV identification approach using CPP features is introduced for the airspace surveillance and air-traffic control. First, the received UAV signal is divided into the overlapped frames, each of which contains a signal subsequence. We propose to apply the energy-to-spectral-entropy ratio to select the frames. For those selected frame, we propose to extract the corresponding CPP features. Then, a lightweight convolutional neural network, namely ResNet-18, is trained to classify the UAV RF signals using the extracted CPPs as the input features for automatic UAV identification. Monte Carlo simulation results justify the superiority of our proposed new UAV identification approach to the existing STFT-based method.

References

[1] M. Stoyanova, Y. Nikoloudakis, S. Panagiotakis, E. Pallis, and E. K. Markakis, "A survey on the internet of things (iot) forensics: Challenges, approaches, and open issues," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 2, pp. 1191–1221, January 2020.

[2] S. A. H. Mohsan, M. A. Khan, F. Noor, I. Ullah, and M. H. Alsharif, "Towards the unmanned aerial vehicles (uavs): A comprehensive review," *Drones*, vol. 6, no. 6, 2022.

[3] K. Belwafi, R. Alkadi, S. A. Alameri, H. A. Hamadi, and A. Shoufan, "Unmanned aerial vehicles' remote identification: A tutorial and survey," *IEEE Access*, vol. 10, pp. 87 577–87 601, August 2022.

[4] A. Jeganathan, B. Dhayabaran, T. D. P. Perera, D. N. K. Jayakody, and P. Muthuchidambaranathan, "Harvest-on-sky: An aoi-driven uav-assisted wireless communication system," *IEEE Internet of Things Magazine*, vol. 5, no. 1, pp. 142–146, March 2022.

[5] Y. Yang, F. Yang, L. Sun, T. Xiang, and P. Lv, "Echoformer: Transformer architecture based on radar echo characteristics for uav detection," *IEEE Sensors Journal*, vol. 23, no. 8, pp. 8639–8653, April 2023.

[6] Y. F. Chen, L. Zhu, C. H. Yao, G. Gui, and L. Yu, "Global context-based threshold strategy for drone identification under the low snr condition," *IEEE Internet of Things Journal*, vol. 10, no. 2, pp. 1332–1346, January 2023.

[7] M. Ezuma, C. K. Anjinappa, V. Semkin, and I. Guvenc, "Comparative analysis of radar-cross-section- based uav recognition techniques," *IEEE Sensors Journal*, vol. 22, no. 18, pp. 17 932–17 949, September 2022.

[8] L. Zhu, S. Zhang, Q. Ma, H. Zhao, S. Chen, and D. Wei, "Classification of uav-to-ground targets based on enhanced micro-doppler features extracted via pca and compressed sensing," *IEEE Sensors Journal*, vol. 20, no. 23, pp. 14 360–14 368, December 2020.

[9] Y. F. Chen, L. Zhu, Y. C. Jiao, C. H. Yao, K. X. Cheng, and Y. T. Gu, "An extreme value theory-based approach for reliable drone rf signal identification," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 2, pp. 454–469, April 2024.

[10] X. Yue, Y. Liu, J. Wang, H. Song, and H. Cao, "Software defined radio and wireless acoustic networking for amateur drone surveillance," *IEEE Communications Magazine*, vol. 56, no. 4, pp. 90–97, April 2018.

[11] G. R. Ding, Q. H. Wu, L. Y. Zhang, Y. Lin, T. A. Tsiftsis, and Y. D. Yao, "An amateur drone surveillance system based on the cognitive internet of things," *IEEE Communications Magazine*, vol. 56, no. 1, pp. 29–35, January 2018.

[12] I. Bisio, C. Garibotto, F. Lavagetto, A. Sciarrone, and S. Zappatore, "Unauthorized amateur uav detection based on wifi statistical fingerprint analysis," *IEEE Communications Magazine*, vol. 56, no. 4, pp. 106–111, April 2018.

[13] X. Yan, P. Yang, X. Zhong, Q. Wang, H. C. Wu, and L. He, "Automatic composite-modulation classification using ultra lightweight deep-learning network based on cyclic-paw-print," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 3, pp. 866–879, June 2024.

[14] M. Guo and Y. Du, "Classification of thyroid ultrasound standard plane images using resnet-18 networks," in *2019 IEEE 13th International Conference on Anti-counterfeiting, Security, and Identification (ASID)*, October 2019, pp. 324–328.

[15] N. N. Yu, S. J. Mao, C. W. Zhou, G. W. Sun, Z. G. Shi, and J. M. Chen, "Dronerfa: A large-scale dataset of drone radio frequency signals for detecting low-altitude drones," *Dianzi Yu Xinxi Xuebao/Journal of Electronics and Information Technology*, vol. 46, no. 4, pp. 1147 – 1156, April 2024.

[16] Y. L. Xie, P. Jiang, Y. Gu, and X. Xiao, "Dual-source detection and identification system based on uav radio frequency signal," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–15, August 2021.

[17] H.-C. Wu, K. Yan, X. Zhang, Y. Wu, and S.-Y. Chang, "Novel robust pilot-tone spotting technique using spectrum clustering," *IEEE Communications Letters*, vol. 23, no. 9, pp. 1590–1593, September 2019.

[18] X. Yan, G. Liu, H. C. Wu, G. Zhang, Q. Wang, and Y. Wu, "Robust modulation classification over α-stable noise using graph-based fractional lower-order cyclic spectrum analysis," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 3, pp. 2836–2849, March 2020.

[19] X. Yan, X. Zhong, H. C. Wu, P. Yang, Q. Wang, and Y. Chen, "Automatic composite-modulation classification using cyclic-paw-print features for cognitive aerospace communications," *IEEE Transactions on Communications*, vol. 72, no. 9, pp. 5486–5502, August 2024.

[20] T. Li, Z. Zhao, C. Sun, R. Yan, and X. Chen, "Adaptive channel weighted cnn with multisensor fusion for condition monitoring of helicopter transmission system," *IEEE Sensors Journal*, vol. 20, no. 15, pp. 8364–8373, August 2020.

[21] M. Akmal, "Tensor factorization and attention-based cnn-lstm deep-learning architecture for improved classification of missing physiological sensors data," *IEEE Sensors Journal*, vol. 23, no. 2, pp. 1286–1294, January 2023.

[22] X. Zhang, X. Zhou, M. Lin, and J. Sun, "Shufflenet: An extremely efficient convolutional neural network for mobile devices," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, December 2018, pp. 6848–6856.

[23] K. Han, Y. H. Wang, H. T. Chen, X. H. Chen, J. Y. Guo, Z. H. Liu, Y. H. Tang, A. Xiao, C. J. Xu, Y. X. Xu, Z. H. Yang, Y. M. Zhang, and D. C. Tao, "A survey on vision transformer," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 1, pp. 87–110, January 2023.